Hamiltonian Boundary Value Method HBVM(4,2)

Rank-two polynomial path with four Gauss nodes

1 Polynomial-path formulation

Let $(c_i, b_i)_{i=1}^4$ be four-point Gauss–Legendre quadrature on $[0, 1]$. For the first two orthonormal shifted Legendre polynomials,

$$P_0(c) = 1, \qquad P_1(c) = \sqrt{3}(2c - 1),$$

define

$$I_{i0} = c_i, \qquad I_{i1} = \sqrt{3}(c_i^2 - c_i).$$

HBVM(4,2) solves for two vector coefficients γ_0, γ_1:

$$Y_i = y_n + h \sum_{j=0}^{1} I_{ij}\gamma_j, \tag{1}$$

$$R_j(\gamma) = \gamma_j - \sum_{i=1}^{4} b_i P_j(c_i) f(t_n + c_i h, Y_i) = 0. \tag{2}$$

The accepted state is

$$y_{n+1} = y_n + h \sum_{i=1}^{4} b_i f(t_n + c_i h, Y_i).$$

Equivalently, the Runge–Kutta matrix is $A = \mathcal{I}_2 \mathcal{P}_2^T \Omega_q$ and has rank two. The nonlinear unknown therefore has twice the state dimension, although four stages evaluate the line integral.

2 Order and energy property

With polynomial degree $s = 2$, the method has classical order $2s = 4$. Four-point Gauss quadrature has degree seven. For an autonomous polynomial Hamiltonian of degree ν, exact polynomial energy conservation follows when $k \geq \nu s/2$; hence HBVM(4,2) covers degrees through four. For a general smooth Hamiltonian the quadrature error makes energy conservation approximate, with an error that decreases rapidly with the step and quadrature accuracy.

HBVM(k, s) is generally not symplectic when $k > s$. Thus HBVM(4,2) must not be labeled symplectic by construction; map-Jacobian diagnostics measure its canonical defect independently of its energy behavior.

3 Nonlinear configurations

Newton acts on the reduced residual (R_0, R_1). The public `jacobian_method` values are `analytic`, `finite_difference`, and `auto`. Analytic mode assembles exact guiding-centre particle blocks at the four stages; finite-difference mode differentiates the stage fields; auto selects the available capability. A backtracking correction is used only when the full Newton update increases the residual. The stopping scale is

$$\varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(1, \|y_n\|_\infty).$$

With `track_energy=True`, the implementation stores the physical Hamiltonian and its drift at saved times. For explicitly time-dependent systems this is a diagnostic, not an invariant claim.

4 Implementation correspondence

Coefficients, residuals, Newton assembly, and integration are in `src/methods/hbvm/order4.py`. The simulation document describes required Hamiltonian capabilities, work counters, fixed-grid sampling, and evaluation studies; shared physical dynamics are documented outside the model tree.

5 Physical accepted states and passive energy tracking

For N particles with d physical components, every accepted snapshot contains $Z \in \mathbb{R}^{dN}$, one scalar time t, and optional separate physical momentum $\kappa \in \mathbb{R}^N$. Thus GC accepted vectors have $2N$ components and FC vectors have $4N$, whether energy is tracked or not. A doubled method prepares temporary $(u, v) \in \mathbb{R}^{2dN}$ copies, allows them to separate, and projects to one physical result before acceptance. The representation stores only the physical size and never owns a clock, momentum, physical coordinate declaration, dynamics or solver.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, Z), \qquad \kappa(t_0) = 0, \qquad \Delta K(t) = H(t, Z(t)) + \kappa(t) - H(t_0, Z_0).$$

GC split-map contributions are summed in temporary work and divided by two to obtain physical momentum. Tracking leaves physical stages, nonlinear stopping scales and adaptive acceptance unchanged. An autonomous Hamiltonian supplies an explicit zero momentum derivative.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired. Dynamics declare component order and position/velocity roles through immutable structural data; initial conditions and results retain those data independently. Methods own maps, projection equations and momentum updates. Generic copy operations live in `src/formulations/state.py`.

Controllers and compiled JAX loops carry physical values and momentum separately. Observers receive physical states. Energy histories in `States.diagnostics` retain their existing labels; `extended_time` has shape (N, n_{samples}) and is broadcast from scalar sample times. Fixed shadow samples leave accepted states and counters unchanged. Adaptive quadrature uses accepted dense output and physical-only error control.

The four converged stage states and Gauss weights integrate the passive momentum. Its value never enters the Legendre coefficient solve.

6 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.